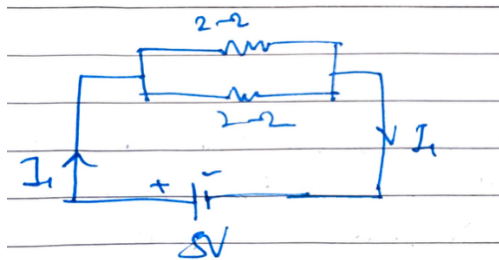


PHYSICS ASSIGNMENT NO 1

Total Marks 10

1. A $12\ \Omega$ resistor is connected to a 24 V supply. Find the current flowing through the resistor. [1 Mark]
2. Find I_1 in the above diagram.



- [The diagram shows two $2\ \Omega$ resistors connected in parallel to a 5 V battery.]. [1 Mark]
3. A resistor of $6\ \Omega$ is connected to a 12 V battery. A charged particle of mass 2 kg experiences a force equal to the current flowing through the resistor. Find its acceleration. [2 Mark]
 4. What is meant by overclocking? [2 Mark]
 5. What is electromagnetism? Explain. [2 Mark]
 6. Which of the following correctly describes the magnetic field near a long straight wire? [1 Mark]
 - a) The field consists of straight lines perpendicular to the wire.
 - b) The field consists of straight lines parallel to the wire.
 - c) The field consists of radial lines originating from the wire.
 - d) The field consists of concentric circles centered on the wire.
 7. At the time of short circuit, the current in the circuit will _____ [1 Mark]
 - a) Reduce sustainably
 - b) Does not change
 - c) Increase heavily
 - d) Vary continuously